

Objective:

This project will focus on designing a novel *Agentic AI Collaboration Insight as a Service* architecture that systematically captures, analyzes, and evaluates collaboration practices among multiple AI agents and their pipelines. The approach brings together insights derived from established agent interaction patterns, coordination mechanisms, and collaboration anti-patterns, providing a holistic view of the quality and effectiveness of agentic AI systems through an “as a service” model.

The proposed concept focuses on assessing how well agentic AI systems are structured, coordinated, and governed, rather than evaluating individual agents in isolation.

Approach:

- **Systematic Review of Agentic Collaboration Practices**
Conduct a comprehensive review of existing collaboration patterns, coordination strategies, and known anti-patterns in agentic AI systems, including multi-agent systems, LLM-based agent frameworks, and autonomous AI workflows.
- **Identification of Key Collaboration Indicators**
Identify the most relevant indicators that characterize effective agent collaboration, such as role clarity, delegation strategies, communication protocols, decision handoff mechanisms, conflict resolution, and human-in-the-loop integration.
- **Design of an Insight-as-a-Service Architecture**
Design an architecture that integrates these collaboration indicators into an actionable service, enabling automated analysis of agentic AI systems and providing structured feedback on collaboration quality, risks, and improvement opportunities.
- **Prototype Implementation**
Implement a prototype of the Agentic AI Collaboration Insight service, capable of analyzing real-world agentic AI pipelines and extracting collaboration-related insights in a scalable and reusable manner.

Requirements:

- Very good knowledge of coding in python and java both for backend and frontend.
- Good knowledge of Machine Learning practices and Reinforcement Learning
- Good knowledge on architecture patterns and practices in software engineering
- Knowledge on CI/CD pipelines and ML/RL pipelines (MLOps/RL Ops)
- Knowledge on Large Language Models (LLMs)